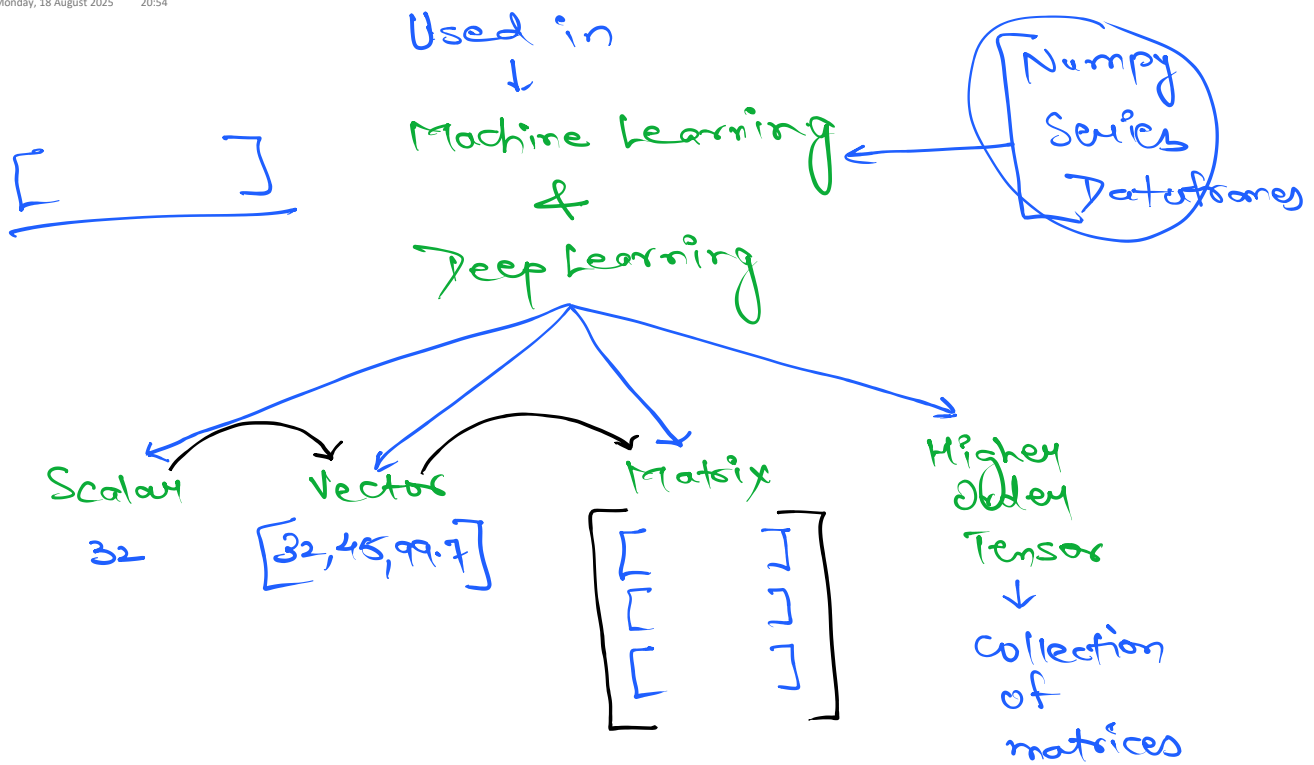




Use of Linear Algebra in Machine Learning →

- ① Represent the data in numerical form so that we can provide that data to machine Learning.

- Text
Audio
Videos
Images
Tabular Data
Signal

Linear Algebra will help us in converting these type of data in numerical form.

Scalar →

34

255

9172.5614

Vectors →

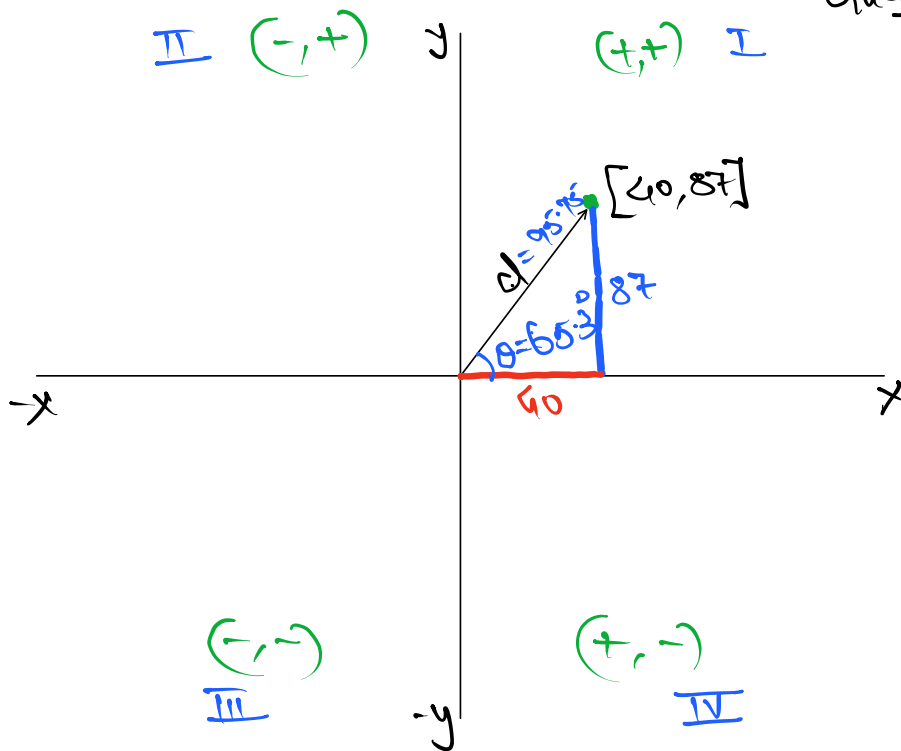
[—————]

1D list

1D Array

Vectors have direction and magnitude.

Quadrants



	x	y
	Profit	Sales
0	$[40]$	$[87]$
1	$[-15]$	$[22]$
2	$[10]$	$[34]$

yes

$[40 \ 87]$

$\begin{bmatrix} 40 \\ 87 \end{bmatrix}$

Magnitude & direction.

Magnitude

$$d^2 = x^2 + y^2$$

$$d = \sqrt{x^2 + y^2}$$
$$= \sqrt{40^2 + 87^2}$$

$$d = 95.75$$

Dissection →

$$\tan \theta = \frac{y}{x}$$

$$\tan \theta = \frac{87}{40}$$

$$\theta = 65.3^\circ$$

Representation of a vector →

- ① Row vectors
- ② Column vectors.

$$[\quad]$$

[]

ML uses column vectors by default.

Ex 7

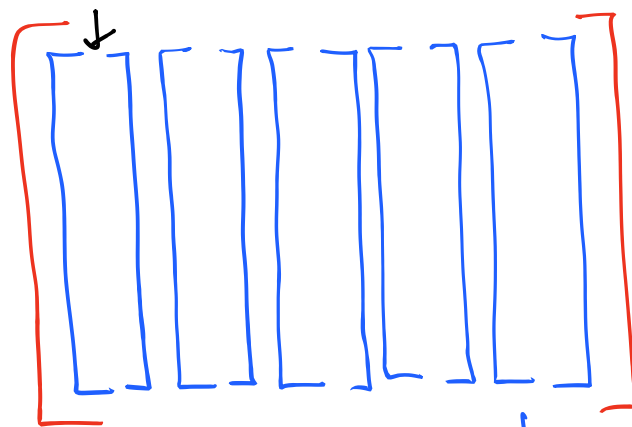
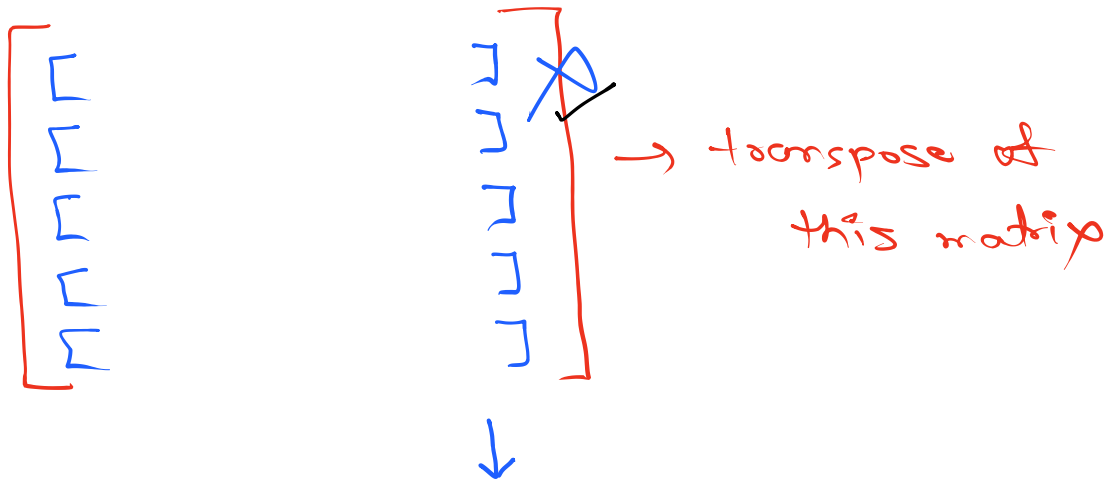
House Price Pred Dataset

	Rooms	Floors	Balcony	Area	Age	Price
0	┌	┌	┌	┌	┌	2.15cr
1	┌	┌	┌	┌	┌	
2	┌	┌	┌	┌	┌	
3	┌	┌	┌	┌	┌	
4	┌	┌	┌	┌	┌	

feature Matrix

House → Hyd → 4, 2, 3, 1800, 2 → ✓

feature matrix

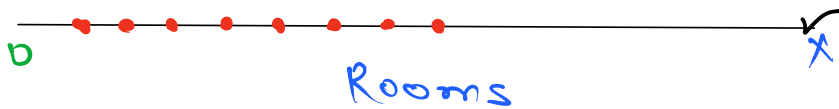


feature Matrix

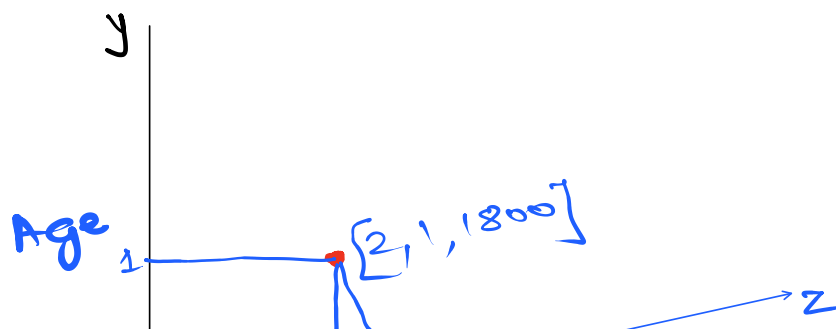
ML

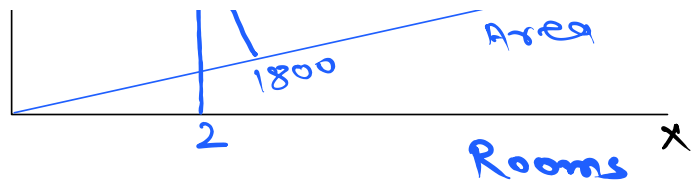
Dimensions of vectors →

Rooms
→ [4]
→ [2]
→ [4]
[6]
[4]
[3]



	Rooms	Age	Area
0	[2]	1	1800]
1	4	3	2200
2	7	2	4200
3	6	3	3800





3d

Linear Algebra

56,877D x